



VIT[®]

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

BCSE204L- Design and Analysis of Algorithms

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Course Objective

1. To provide mathematical foundations for analysing the complexity of the algorithms
2. To impart the knowledge on various design strategies that can help in solving the real world problems effectively
3. To synthesize efficient algorithms in various engineering design situations



BCSE204L- Design and Analysis of Algorithms- Course outcome

1. Apply the mathematical tools to analyse and derive the running time of the algorithms
2. Demonstrate the major algorithm design paradigms.
3. Explain major graph algorithms, string matching and geometric algorithms along with their analysis.
4. Articulating Randomized Algorithms.
5. Explain the hardness of real-world problems with respect to algorithmic efficiency and learning to cope with it.



BCSE204L- Design and Analysis of Algorithms- Syllabus

Module:1	Design Paradigms: Greedy, Divide and Conquer Techniques	6 hours
Overview and Importance of Algorithms - Stages of algorithm development: Describing the problem, Identifying a suitable technique, Design of an algorithm, Derive Time Complexity, Proof of Correctness of the algorithm, Illustration of Design Stages - Greedy techniques: Fractional Knapsack Problem, and Huffman coding - Divide and Conquer: Maximum Subarray, Karatsuba faster integer multiplication algorithm.		
Module:2	Design Paradigms: Dynamic Programming, Backtracking and Branch & Bound Techniques	10 hours
Dynamic programming: Assembly Line Scheduling, Matrix Chain Multiplication, Longest Common Subsequence, 0-1 Knapsack, TSP- Backtracking: N-Queens problem, Subset Sum, Graph Coloring- Branch & Bound: LIFO-BB and FIFO BB methods: Job Selection problem, 0-1 Knapsack Problem		
Module:3	String Matching Algorithms	5 hours
Naïve String-matching Algorithms, KMP algorithm, Rabin-Karp Algorithm, Suffix Trees.		
Module:4	Graph Algorithms	6 hours
All pair shortest path: Bellman Ford Algorithm, Floyd-Warshall Algorithm - Network Flows: Flow Networks, Maximum Flows: Ford-Fulkerson, Edmond-Karp, Push Re-label Algorithm – Application of Max Flow to maximum matching problem		
Module:5	Geometric Algorithms	4 hours
Line Segments: Properties, Intersection, sweeping lines - Convex Hull finding algorithms: Graham's Scan, <u>Jarvis' March Algorithm</u> .		
Module:6	Randomized algorithms	5 hours
Randomized quick sort - The hiring problem - Finding the global Minimum Cut.		
Module:7	Classes of Complexity and Approximation Algorithms	7 hours
The Class P - The Class NP - Reducibility and NP-completeness – SAT (Problem Definition and statement), 3SAT, Independent Set, Clique, Approximation Algorithm – Vertex Cover, Set Cover and Travelling salesman		
Module:8	Contemporary Issues	2 hours



BCSE204L- Design and Analysis of Algorithms- References

Text Books:

1. Thomas H. Cormen, C.E. Leiserson, R L. Rivest and C. Stein, Introduction to Algorithms, 2009, 3rd Edition, MIT Press.

Reference Books:

1. Jon Kleinberg, Éva Tardos ,Algorithm Design, Pearson education, 2014.
2. Rajeev Motwani, Prabhakar Raghavan; Randomized Algorithms, Cambridge University Press, 1995 (Online Print – 2013)
3. Ravindra K.Ahuja, Thomas L.Magnanti and James B.Orlin, “ Network Flows: Theory, Algorithms and Applications”, Pearson Education, 2014.



Module 5 : Geometric Algorithms (4 Hours)

- Line Segments, Properties, Intersection, Sweeping Lines
- **Convex hull finding Algorithm**
- Graham's Scan
- **Jarvis March Algorithm**



Some Terminologies

- **Polygon:** Curve ending on itself that is formed by a sequence of straight-line segments, called the sides of the polygon.
- **Convex Set:** A set of points on plane is called convex if for any 2 points P and Q, the entire line segments with end points P and Q belongs to the set.
- **Vertex:** A point joining two consecutive sides is a of the polygon.
- **Boundary-** Set of points on the polygon itself.
- **Interior-** Set of points in the plane enclosed by a Simple polygon.
- **Exterior-** set of points surrounding polygon.
- **Convex Polygon-** Given any two points on the boundary or interior of a simple polygon, all points on the line segment drawn between them are contained in the polygon's boundary or interior.



Convex Hull



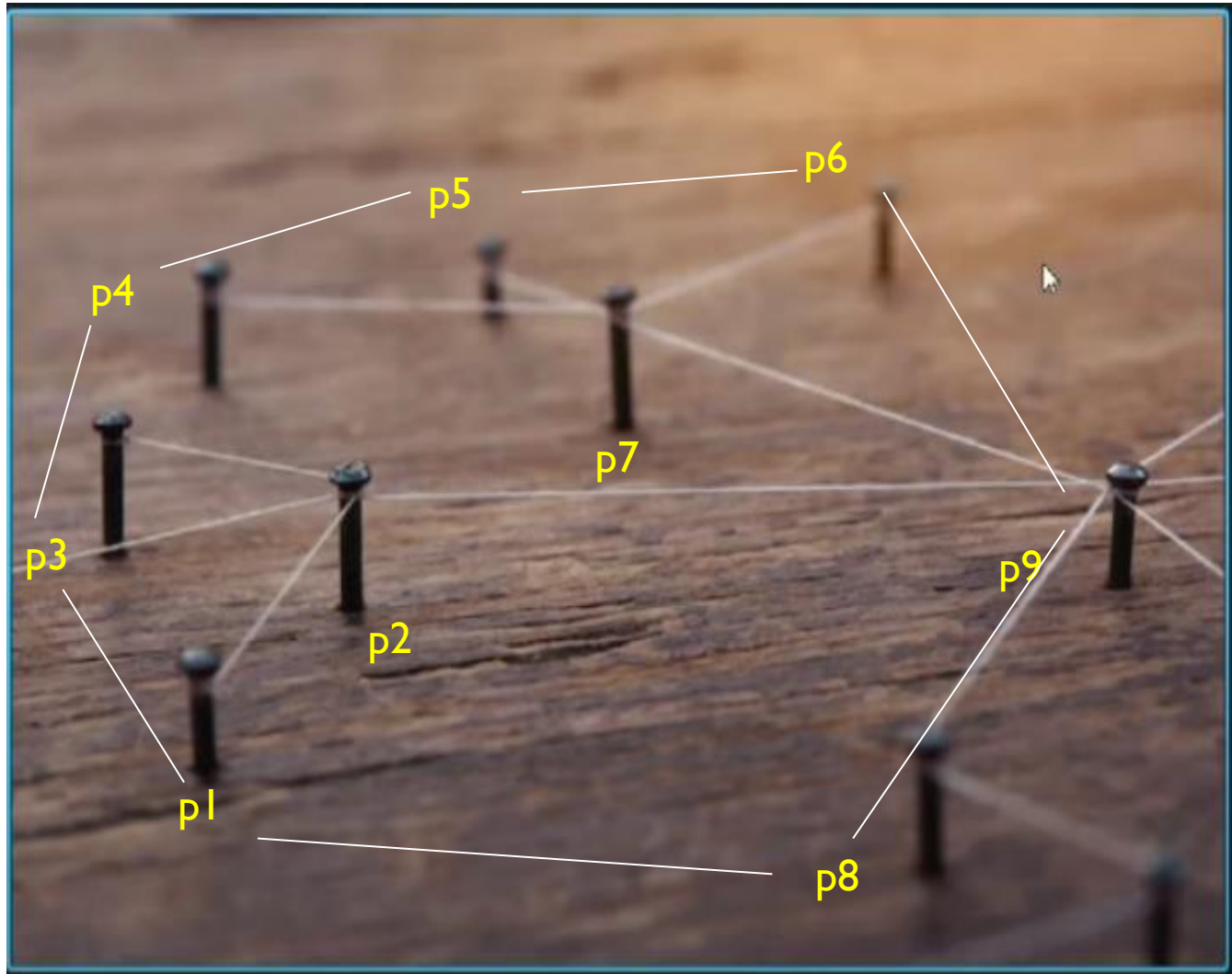
Problem Statement: Having set of points , **our objective is to draw the convex polygon**, How it should be?? - All points should be covered – either inside polygon or in the boundary.

Note: No .of points ≥ 3 and points should not be in collinear





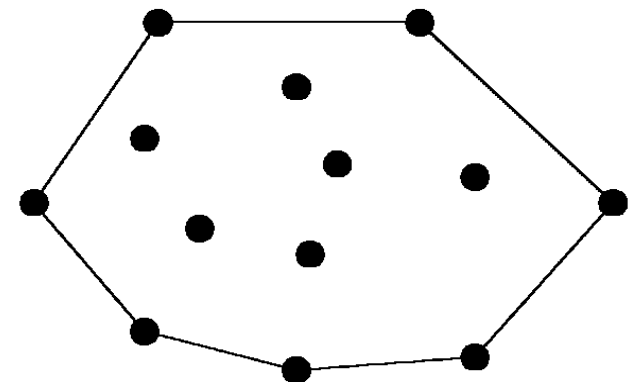
Convex Hull





Convex Hull- Definition

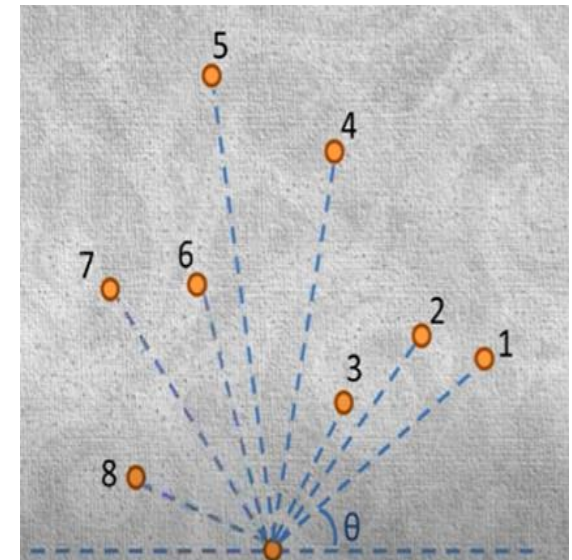
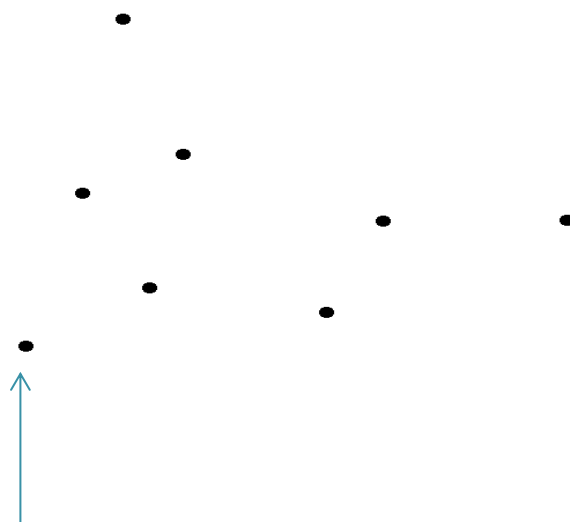
- The convex hull of a set Q of points, denoted by $CH(Q)$ is the smallest convex polygon P for which each point in Q is either on the boundary of P or in its interior.
- **Note :** All points in the set Q are unique and that Q contains at least three points which are not colinear.
- Vertices maximizes the area while minimizing the circumference.



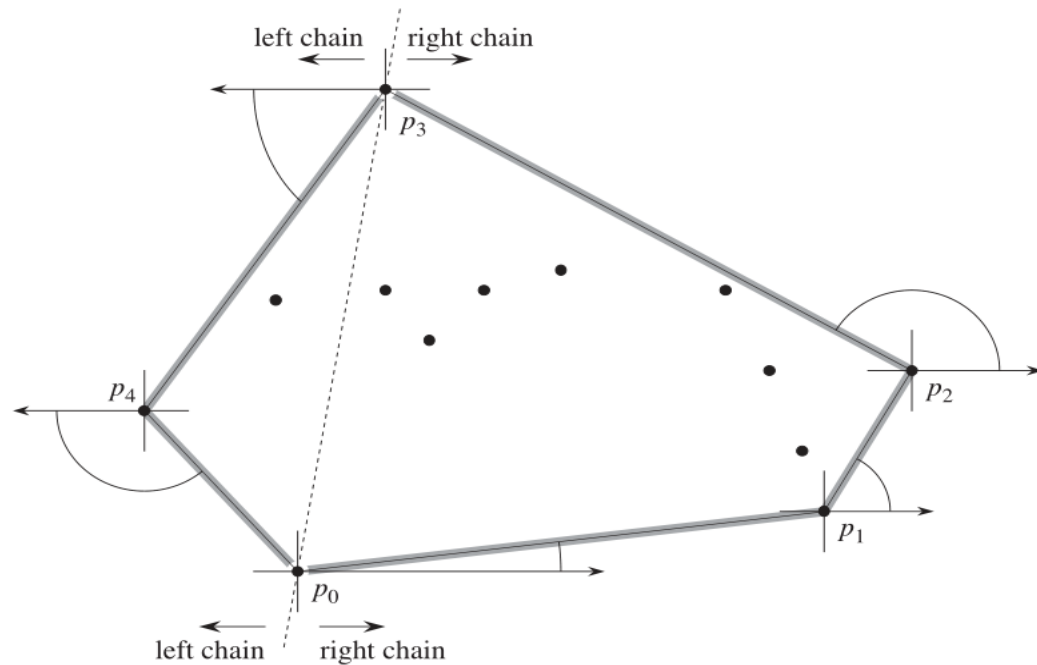


Jarvis's March / Gift Wrapping algorithm

- Computes the convex hull of a set Q of points by a technique known as package wrapping (or gift wrapping).
- Algorithm runs in $O(n h)$ time.
- $h \rightarrow$ number of vertices of $CH(Q)$



Jarvis's March

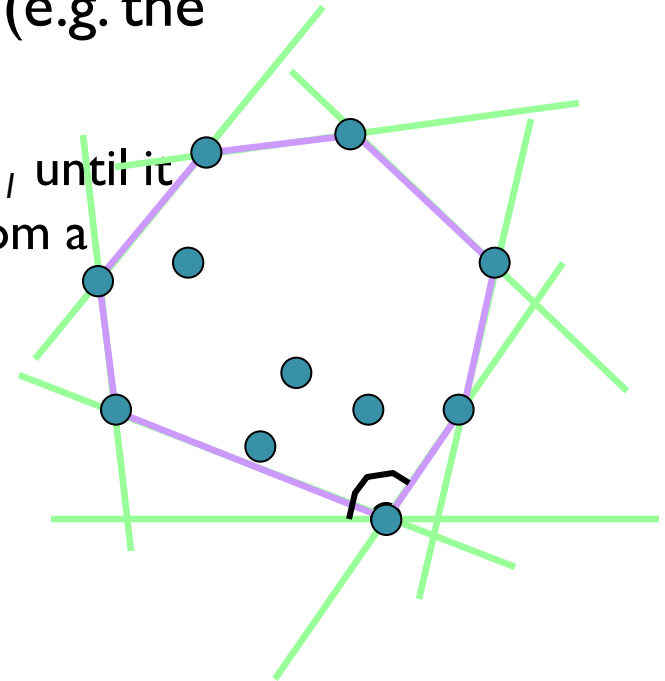


- Choose the first vertex as the lowest point p_0
- Next vertex p_1 , has the smallest polar angle of any point with respect to p_0 , Then, p_2 has the smallest polar angle with respect to p_1 .
- Right chain goes as high as the highest point p_3 .
- Then, we construct the left chain by finding smallest polar angles with respect to the **negative x-axis**.



Jarvis's March

- Algorithm:
- Find a point p_1 on the convex hull (e.g. the lowest point).
- Rotate counterclockwise a line through p_1 until it touches one of the other points (start from a horizontal orientation).
- Question: How is this done ?**



- Repeat the last step for the new point.
- Stop when p_1 is reached again.

Time Complexity: $O(nh)$, where n is the input size and h is the output (hull) size.



Thank you